Experiment No. 05

Aim:

To apply software metrics (LOC, Function Points, Productivity, COCOMO) to estimate the effort, time, and cost.

Topic name: Employee Management System

Objective:

1. To understand the role of software metrics in project estimation.

2. To apply LOC, Function Point, Productivity, and COCOMO techniques for effort and time calculation.

3. To estimate cost and resources required for a software project.

Step 1: Identify Metrics

Decide which metric to use for estimation:

1. LOC (Lines of Code) – derived from FP using the assumed productivity.

2. Function Points (FP) – based on inputs, outputs, files, inquiries, interfaces.

3. COCOMO Model – uses LOC + Salary per month.

Step 2: Collect Basic Data

For **Employee Management System:**

- **External Inputs (EI):** Admin Login, Employee Registration, Add/Update/Delete Employee, Department Management, Attendance, Leave, Salary → **10**
- **External Outputs (EO):** Employee Report, Attendance Report, Leave Report, Salary Report, Department Report → **5**
- **External Inquiries (EQ):** Search Employee, View Employee Details, Check Attendance, Check Leave Status → **4**
- **Internal Logical Files (ILF):** Employee DB, Department DB, Attendance DB, Leave DB, Salary DB, Admin DB → **6**
- **External Interface Files (EIF):** Email/Notification Service, Payroll/Banking Service → **2**

Step 3: Calculate Size (FP / LOC)

Use Average weights (if detailed complexity not rated):

EI = 4, EO = 5, EQ = 4, ILF = 10, EIF = 7

Unadjusted FP (UFP):

For the **Employee Management System**:

- EI = 10
- EO = 5
- EQ = 4
- ILF = 6
- EIF = 2

UFP = (10×4) + (5×5) + (4×4) + (6×10) + (2×7)

UFP = 40 + 25 + 16 + 60 + 14

UFP = 155

Value Adjustment Factor (VAF): 0.65+(0.01×Total Degree of Influence (DI))

Where **DI** = sum of ratings (0–5) across 14 General System Characteristics

No.	GSC	Rating (0–5)	Reason
1	Data communications	3	System communicates with database and notification services.
2	Distributed data processing	2	Mainly centralized database with limited distributed processing.
3	Performance	3	Employee, attendance and salary records require good response time.
4	Heavily used configuration	3	Multiple administrators/users may access the system.
5	Transaction rate	3	Medium transaction load for attendance, leave and salary updates.
6	Online data entry	4	Employee and attendance information is frequently entered online.
7	End-user efficiency	4	Dashboard and simple interfaces are required for efficient operation.
8	Online update	4	Employee, attendance, leave and salary data are updated online.
9	Complex processing	3	Salary calculation, attendance and leave processing have medium complexity.

10	Reusability	2	Some modules can be reused, but the system is domain-specific.
11	Installation ease	2	Web-based system requires moderate server/database setup.
12	Operational ease	3	System should be simple for administrators and staff to operate.
13	Multiple sites	2	The system may support employees from different departments/locations.
14	Facilitate change	3	Employee policies, departments and salary structures may change.

Total DI Calculation

DI = 3+2+3+3+3+4+4+4+3+2+2+3+2+3

DI = 43

Calculate VAF

VAF = 0.65 + (0.01 × 43)

VAF = 0.65 + 0.43

VAF = 1.08

Final FP (After VAF)

FP = UFP × VAF

FP = 155 × 1.08

FP = 167.4 ≈ 167 FP

Language Ratio and Estimated LOC

Assuming the Employee Management System is developed using Java:

Language ratio: Java ≈ 53 LOC/FP

Estimated LOC

LOC = FP × LOC/FP

LOC = 167 × 53

LOC = 8,851 LOC

Estimated Size ≈ 8.85 KLOC

Step 4: Convert Size to Effort, Time & Cost

A) Productivity Approach (Simple)

- Assume productivity = 500 LOC / Person Month (PM)
- Effort = LOC / Productivity

Effort = 8,851 / 500

Effort = 17.70 PM

B) Basic COCOMO (Semi-Detached – Typical Team and Mixed Experience)

Constants:

- a1 = 3.0
- a2 = 1.12
- b1 = 2.5
- b2 = 0.35

Size: KLOC = 8.851

Effort (PM)

Effort = a1 × (KLOC)^a2

Effort = 3.0 × (8.851)^1.12

Effort = 34.49 PM

Development Time

Time (months) = b1 × (Effort)^b2

Time (months) = 2.5 × (34.49)^0.35

Time (months) = **8.63 months**

Team Size

Team Size = Effort / Development Time

Team Size = 34.49 / 8.63

Team Size = ≈ 4 persons

Cost Estimation

Assume salary per person-month = **₹50,000**

Total Cost = 34.49 × ₹50,000

Total Cost = ₹17,24,500 approximately

Step 5: Present Results

"Estimated effort ≈ **34.49 PM**, duration ≈ **8.63 months**, team size ≈ **4 persons**, and total cost ≈ **₹17.25 lakh** for the **Employee Management System**."